1. A proton is accelerated to 0.999c. (a) What is the proton's momentum? (b) By what factor does the proton's momentum exeed its Newtonian Momentum?

2. A 1.0g particle has momentum 400,000 kg m/s. What is the particle's speed in m/s?

3. At what speed, as a fraction of c, is a particle's momentum twice its Newtonian value?

4. Show, using the binomial approximation (i.e that $(1+x)^2 \approx 1+nx$ for very small x) that relativistic Kinetic Energy $KE = (\gamma - 1)\, mc^2$ reduces to the Newtonian $KE = \frac{1}{2}mv^2$ in the low velocity limit.

5. An electron of rest mass $m_e = 9.11 \times 10^{-31}$ kg moves through a laboratory with $v = \frac{2}{3}c$. Use a Mathematica script to:

- **A.** Compute E_0, E, p_x and KE.
- **B.** Compute the momentum 4-vector
- **C.** A second reference frame, S', shoots by with $\beta = 0.99$. Use mathematica to compute the components of the momentum 4-vector in S'

6. Four-vector A has components $A^t = 3L$ and $A^x = 2L$. It is known that the invariant of A is L_0. Find an expression for L in terms of L_0.